

Differentiability

If $y = f(x)$, then $\frac{d}{dx} \{f(x)\} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Now, the function will be differentiable at $x=c$.

$$\text{If } L.H.D = R.H.D$$

when $L.H.D = Lf'(c)$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(c)}{h}$$

and $R.H.D = Rf'(c)$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(c)}{h}$$

Q. Test The differentiability of the function

$$f(x) = \begin{cases} 5x-4, & 0 \leq x < 1 \\ 4x^2-3x, & 1 \leq x < 2 \end{cases} \text{ at the point } x=1$$

Soln: Given that.

$$f(x) = \begin{cases} 5x-4, & 0 \leq x < 1 \\ 4x^2-3x, & 1 \leq x < 2 \end{cases}$$

Now, $L.H.D = Lf'(c)$

$$= \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(c)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5(1+h) - 4 - (5 \cdot 1 - 4)}{h}$$

[এখানে $5x-4$ ব্যৱহাৰ কৰা
কৰ্মক্ষেত্ৰৰ কাৰণত এটা সৰ্বস্বত্ব
প্ৰদান কৰা হৈছে]

$$= \lim_{h \rightarrow 0} \frac{5 + 5h - 4 - 1}{h}$$

D.H.R. = D.H.L.

(৩) $f(x) = f(a)$ নতুন

$$= \lim_{h \rightarrow 0} \frac{5h}{h}$$

$$\frac{(x)f - (a)f}{x-a} \text{ mit } \frac{0}{0} = \frac{f'(a)}{1}$$

$$= \lim_{h \rightarrow 0} 5 = 5$$

(৩) $f(x) = f(a)$ and R.H.D. = R.H.L.

$$\frac{(x)f - (a)f}{x-a} \text{ mit } \frac{0}{0} = \frac{f'(a)}{1}$$

Again

R.H.D. = R.F.(c) Differentiation test - 2

$$= \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4(1+h)^2 - 3(1+h) - (5 \cdot 1 - 4)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4 + 8h + 4h^2 - 3 - 3h - 1}{h}$$

$$\lim_{h \rightarrow 0^+} \frac{h(4h+5)}{h}$$

$$\lim_{h \rightarrow 0^+} 5 = 5$$

As, L.H.S = R.H.D.

So, the given equation is differentiable at $x=1$.

Homework

Q2 Types Q2 2nd Math 2018-19

Indeterminate (Form)

Q. Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2\sin x - 4x}{x^2}$

→ Soln: Given the limit.

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2\sin x - 4x}{x^2}$$

As, $\frac{0}{0}$ Form, using L'Hospital rule

we get, $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2\cos x - 4}{2x}$

Again, $\frac{0}{0}$ form. So, using L'Hospital Rule,

$$\text{we get } \lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x}{2}$$

$$\frac{e^0 - e^{-0} + 2 \sin 0}{2}$$

$$\frac{1 - 1 + 0}{2} = 0$$

Homework

2) $\lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2}$

3) $\lim_{\theta \rightarrow \pi/4} \frac{x\sqrt{2} - \cos\theta - \sin\theta}{(4\theta - \pi)^2}$

Q → Evaluate the limit $\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$ using L'Hospital rule.

→ Given the limit

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$$

As ∞^0 , so, let.

$$y = \lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$$

Taking $\ln$ on both sides - we get,

$$\ln y = \lim_{x \rightarrow 0} \tan x \ln \frac{1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\cot x} \ln x^{-1}$$

$$= \lim_{x \rightarrow 0} \frac{-\ln x}{\cot x}$$

As $\frac{0}{0}$ Form. So using L'Hospital rule.

we get,

$$\ln y = \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{-\operatorname{cosec}^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x}$$

As $\frac{0}{0}$ Form, so, using L'Hospital rule.

we get,

$$\ln y = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{1}$$

$$= 2 \sin 0 \cdot \cos 0 = 0$$

on $e^{\ln y}$, e^0

or, $y = 1$.

Hence $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x} = 1$, that is at $x=0$, the limit is 1.

Homework

2) $\lim_{x \rightarrow 0} x^2 \log x^2$

3) $\lim_{x \rightarrow 0} (e^x + 1)^{-2/x}$

Definition

1. Function
2. Domain
3. Range
4. limit
5. continuity
6. Differentiability
7. Leibniz theorem
8. rule's theorem
9. Min-value Theorem.
10. Maclaurin

11. Taylor Theorem.